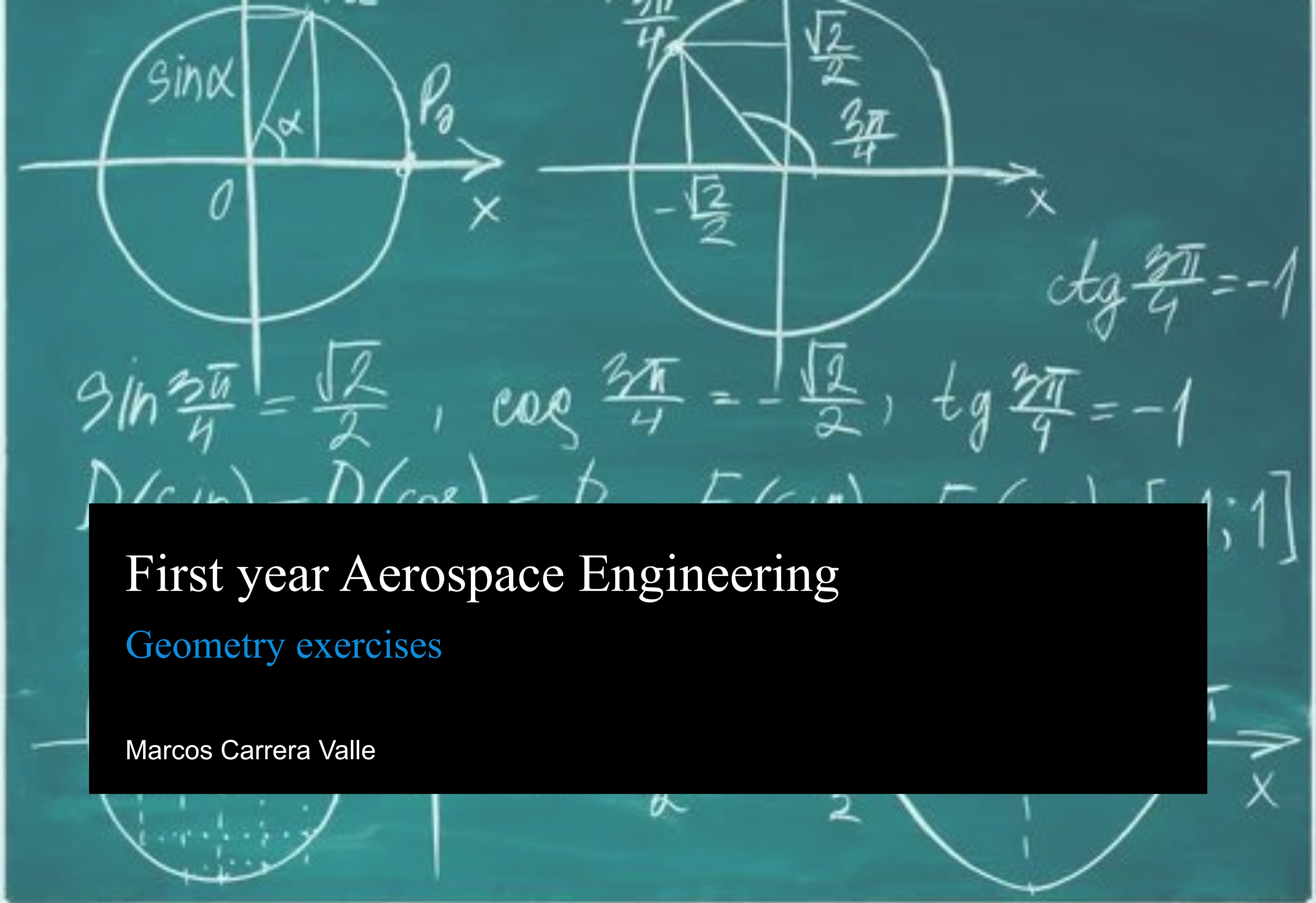

First year Aerospace Engineering

Geometry exercises

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1) Find the area of a scalene triangle with side lengths 4 cm, 13 cm, and 15 cm.
2) Find the volume of a cone with a radius of 3 units and a height of 12 units (leave answer in terms of π).
3) Find the perimeter of a regular heptagon with a side length of 9 cm.
4) Find the total area of a composite 2D shape consisting of a square (side length 4) with an equilateral triangle attached to the top side.
5) Find the total surface area of a square-based pyramid with a base side length of 10 units and a triangular face slant height of 8 units.
6) Find the volume of a composite solid made of a cylinder (radius 3, height 4) with a hemisphere of the same radius placed on top (leave answer in terms of π).
7) Find the area of a regular hexagon with a side length of 2 units.
8) Find the area of an isosceles trapezoid with parallel bases of length 8 and 14, and a height of 4.
9) Find the volume of a triangular prism where the base is a triangle with area 12 cm^2 and the length of the prism is 10 cm.
10) Find the area of a sector of a circle with a radius of 6 units and a central angle of 60 degrees.

11) A triangle has two sides of length 5 and 12. The angle between them is 120 degrees. Find the exact length of the third side.
12) A triangle has sides of length 7, 8, and 9. Find the cosine of the largest angle (leave as a fraction).
13) A triangle has sides of length 13, 14, and 15. Calculate the exact area of the triangle.
14) In triangle ABC, angle A is 30 degrees, side a (opposite A) is 10, and side b (adjacent to A) is 16. There are two possible values for angle B. Find sin(B).
15) A triangle has an area of 30. Two of its sides are 5 and 13. Find the angle included between these two sides.
16) In triangle ABC, angle A is 45 degrees, angle B is 60 degrees, and side a is 20. Find the exact length of side b.
17) A triangle has sides 20, 21, and 29. Verify if it is a right-angled triangle, and find the length of the adjacent
18) A triangle has sides 5, 5, and 6. First find the are, then use that area to find the height of the triangle corresponding to the base of length 6.
19) A triangle has sides 6 and 10, and the angle made by these sides is 150 degrees. Find the area of this triangle.
20) A triangle has sides 4, 6, and 8. Find the length of the median drawn to the longest side.

Find the equation of the tangent line to the following functions at the specified point:

21) $f(x) = x * \mathrm{sqrt}(x^2 + 3)$ at $x = 1$

22) $f(x) = (x + 1) / (x - 2)$ at $x = 3$

23) $f(x) = (x^2 - 3)^5$ at $x = 2$

24) $f(x) = x^2 * \ln(x)$ at $x = e$

25) $f(x) = e^{(\sin x)}$ at $x = \pi$

26) $f(x) = \cos(x) / (1 + \sin x)$ at $x = 0$

27) $f(x) = (\sin x)^2$ at $x = \pi/4$

28) $f(x) = (3x + 1) * e^{(2x)}$ at $x = 0$

29) $f(x) = \mathrm{sqrt}(4x / (x + 3))$ at $x = 1$

30) $f(x) = \tan(2x)$ at $x = \pi/8$

Find the equation of the tangent line to the following functions at the specified point:

31) Find the point on the curve $y = \sqrt{x}$ that is closest to the point (2, 0), and then calculate the distance.

32) Find the point on the parabola $y = x^2$ that is closest to the point (0, 5), and then calculate the distance.

33) Find the point on the parabola $y = x^2$ that is closest to the line $y = 2x - 5$, and then calculate the shortest distance.

34) Find the point on the curve $y = e^x$ that is closest to the line $y = x - 2$, and then calculate the shortest distance.

35) Find the point on the curve $y = 1/x$ (for $x > 0$) that is closest to the line $y = -x - 4$, and then calculate the shortest distance.

Find the equation of the tangent line to the following functions at the specified point:

36) Find the point on the curve $y = e^x$ closest to the curve $y = \ln(x)$, and then calculate the shortest distance between the two curves.

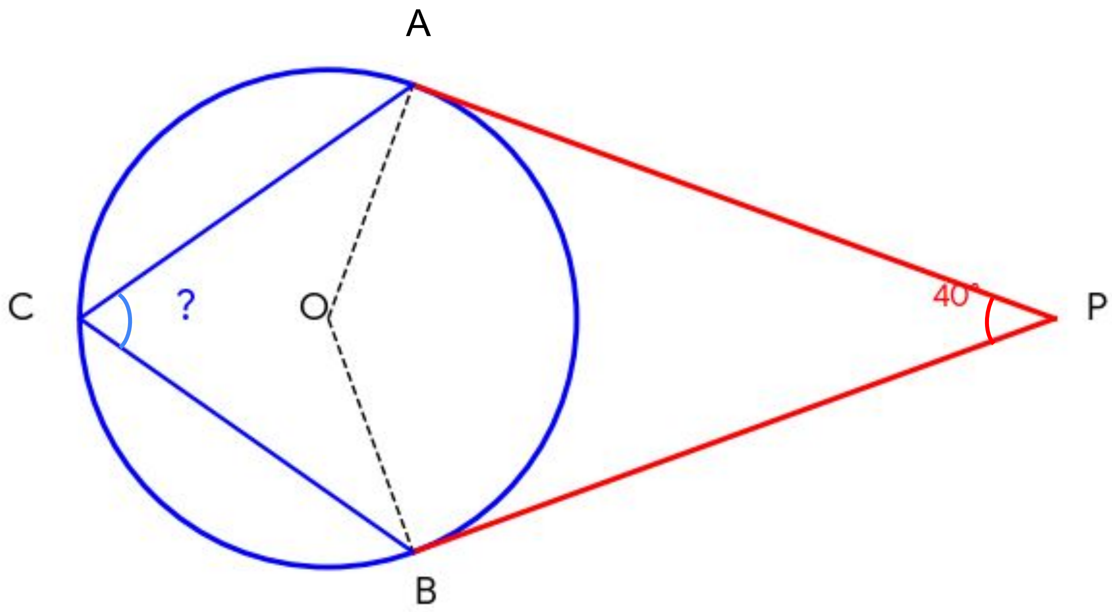
37) Find the point on the parabola $y = x^2 + 1$ closest to the parabola $y = -x^2 - 1$, and then calculate the shortest distance.

38) Find the point on the circle $x^2 + y^2 = 1$ closest to the circle $(x - 3)^2 + (y - 4)^2 = 1$, and then calculate the shortest distance.

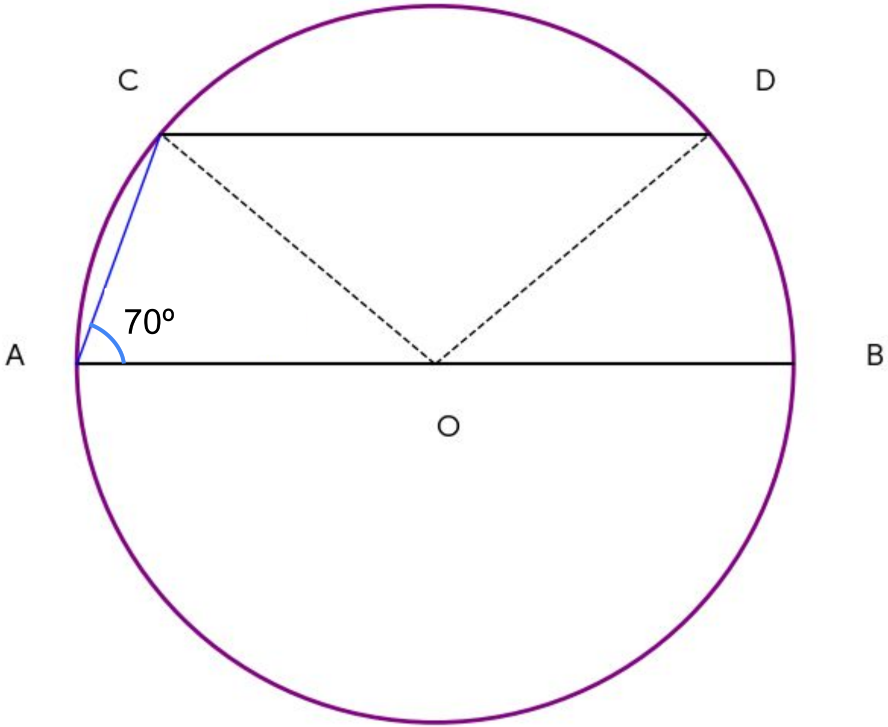
39) Find the point on the parabola $y = x^2 + 4$ closest to the circle $x^2 + (y - 1)^2 = 1$, and then calculate the shortest distance.

40) Find the point on the hyperbola $xy = 8$ (for $x > 0$) closest to the circle $x^2 + y^2 = 2$, and then calculate the shortest distance.

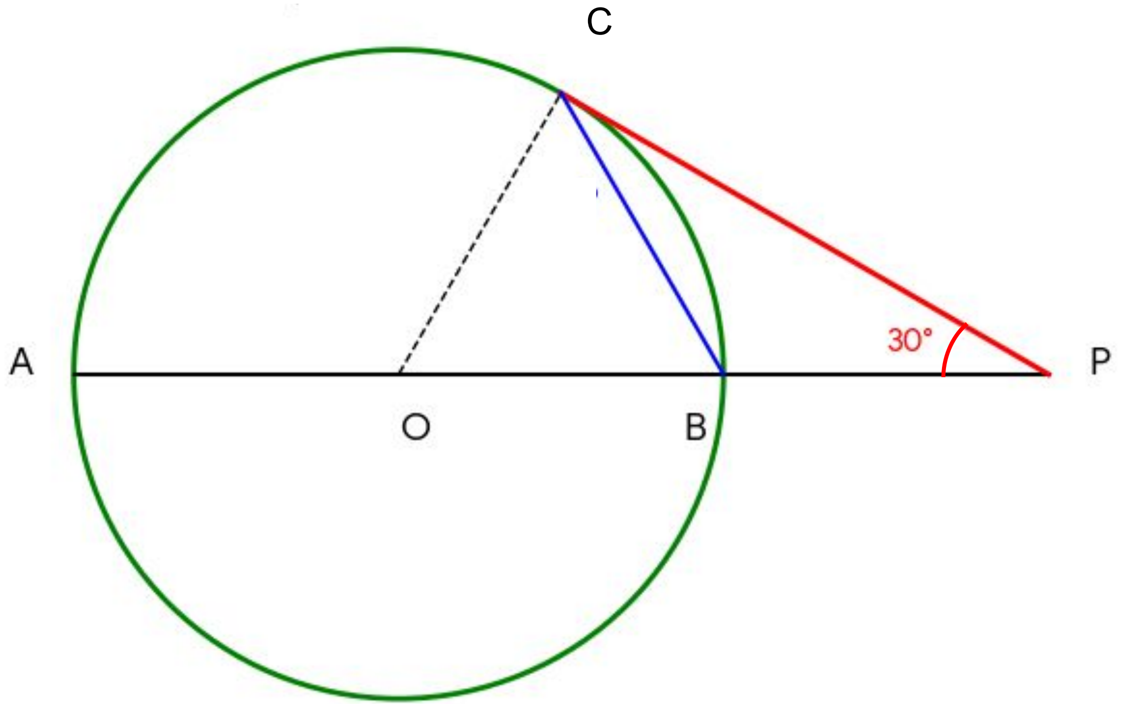
41) Find angle ACB



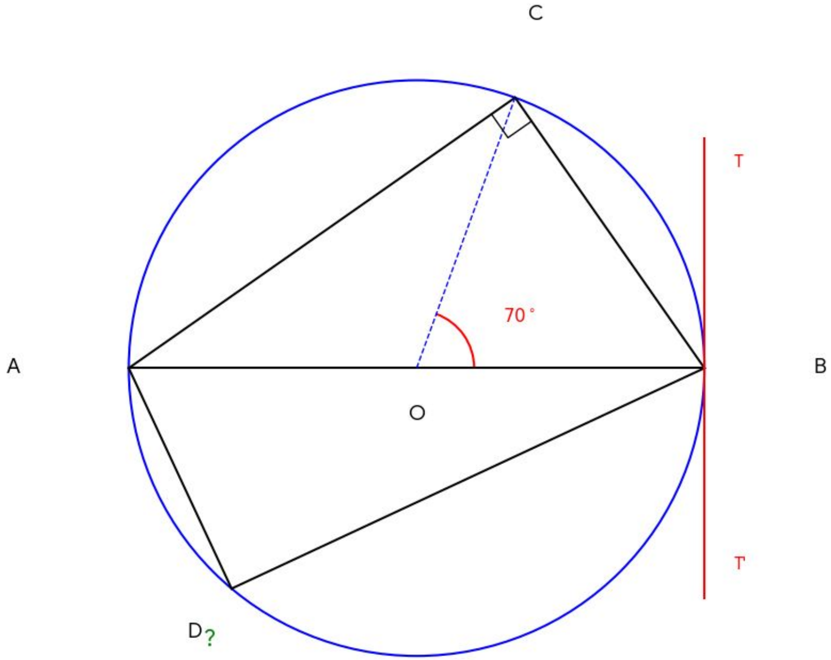
42) Find angle COD



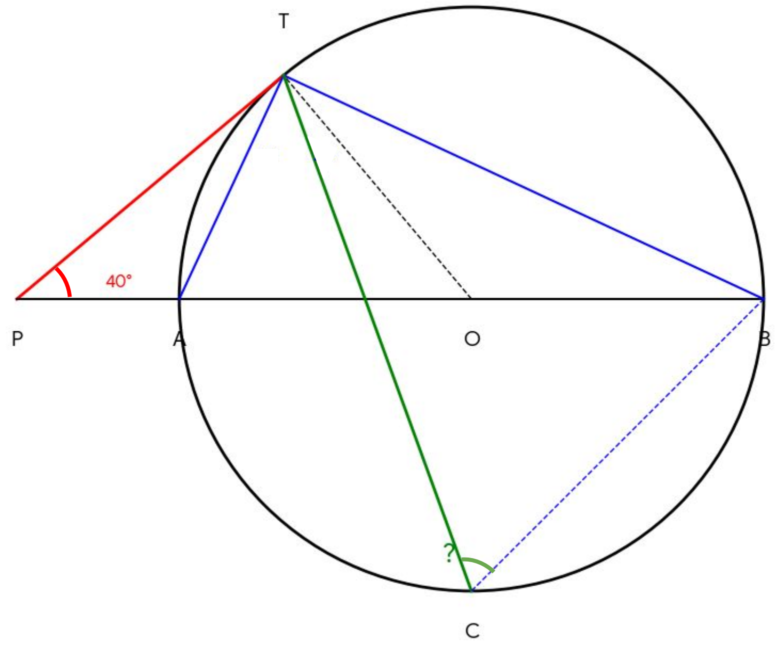
43) Find angle PCB



44) Find angle ADC



45) Find angle TCB



SOLUTIONS

1) 24 cm^2
2) 36 * pi cubic units
3) 63 cm
4) 16 + 4 * sqrt(3) square units
5) 260 square units
6) 54 * pi cubic units
7) 6 * sqrt(3) square units
8) 44 square units
9) 120 cm^3
10) 6 * pi square units

11) sqrt(229)
12) 2/7
13) 84
14) 0.8
15) 67º
16) 10 * sqrt(6)
17) Adjacent = 420/29
18) Area = 12, Height = 4
19) 15
20) sqrt(10)

Find the equation of the tangent line to the following functions at the specified point:

21) $y = 1.5x + 0.5$ (or $y = 3/2\ x + 1/2$)

22) $y = -3x + 13$

23) $y = 20x - 39$

24) $y = (3e)x - 2e^2$

25) $y = -x + \pi + 1$

26) $y = -x + 1$

27) $y = x + 0.5 - \pi/4$

28) $y = 5x + 1$

29) $y = 0.375x + 0.625$ (or $y = 3/8\ x + 5/8$)

30) $y = 4x + 1 - \pi/2$

Find the equation of the tangent line to the following functions at the specified point:

31) Point (1.5, 1.22) (Exact: (1.5, sqrt(1.5))); Distance: sqrt(7) / 2

32) Point (2.12, 4.5) (Exact: (sqrt(4.5), 4.5)); Distance: sqrt(19) / 2

33) Point (1, 1); Distance: 4 / sqrt(5)

34) Point (0, 1); Distance: 3 / sqrt(2)

35) Point (1, 1); Distance: 3 * sqrt(2)

Find the equation of the tangent line to the following functions at the specified point:

36) Point (0, 1) (on e^x); Distance: sqrt(2)

37) Point (0, 1) (on first parabola); Distance: 2

38) Point (0.6, 0.8) (on first circle); Distance: 3

39) Point (0, 4) (on parabola); Distance: 2

40) Point (2.83, 2.83) (Exact: (2 * sqrt(2), 2 * sqrt(2))); Distance: 4 - sqrt(2)

41) 70º

42) 110º

43) 30º

44) 125º

45) 65º